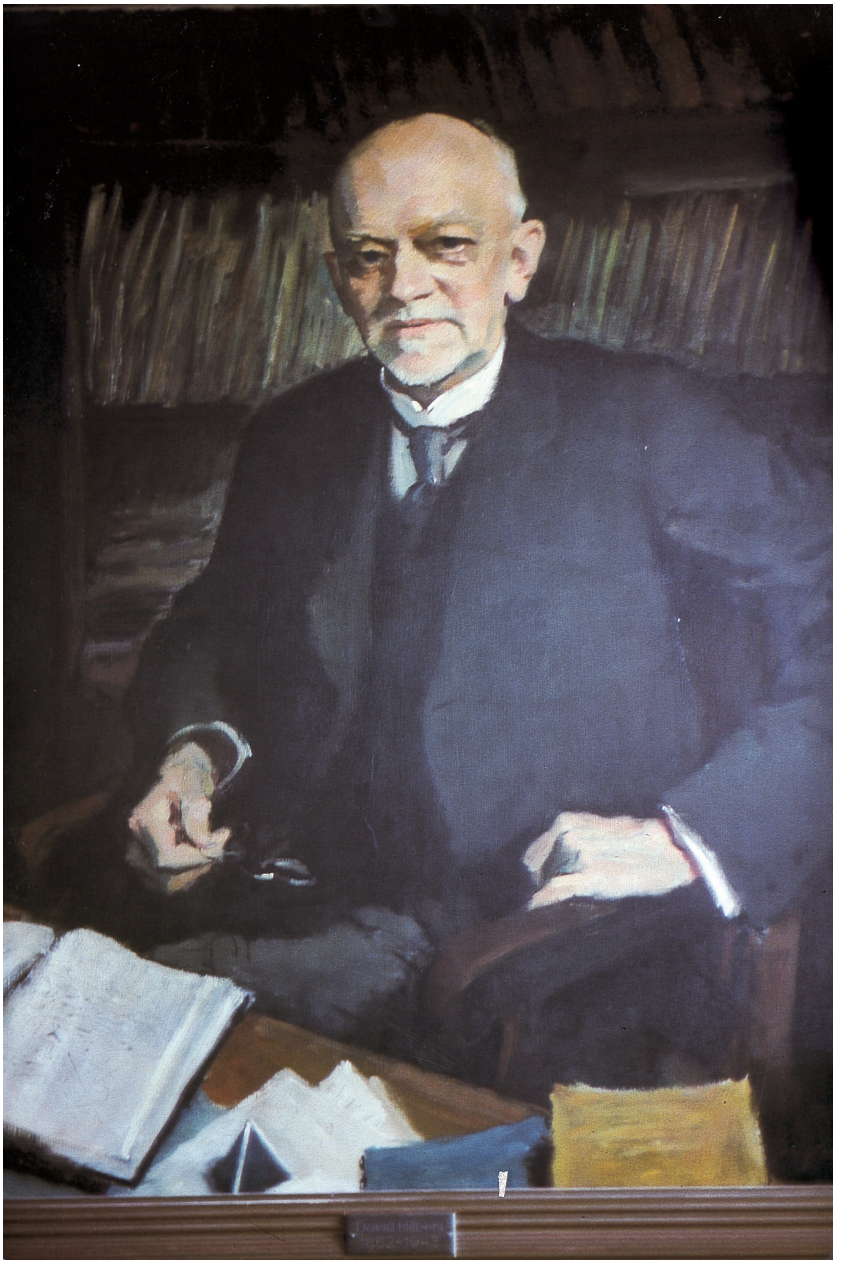
Part V

Hilbert



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David Hilbert (1862–1943) grew up in Königsberg, then capital of East Prussia and now Kaliningrad in Russia, in an upper middle-class family; his father was a judge. The town had been home to the philosopher Immanuel Kant, to Leonard Euler, whose solution to the riddle of how to cross its seven bridges across the river Pregel without re-using one became a starting point for graph theory and topology, and to C. G. J. Jacobi.

After an unimpressive school career, he studied at the university to graduate with his doctoral thesis on invariant theory in 1885. He worked in this area until 1893, proving among other things the *Hilbert basis theorem* saying that any ideal in a polynomial ring (in finitely many variables over a field) is finitely generated (Theorem 21.23), and introducing the *Hilbert function* of algebraic varieties.

Two further results from his "multivariate polynomial phase" are relevant to the subject matter of this text: firstly *Hilbert's Nullstellensatz*[1] (1890), which says that if a polynomial g vanishes on the set of common roots of some multivariate polynomials $f_1, \ldots, f_s$ over $\mathbb{C}$, then some power g^e is in the ideal $\langle f_1, \ldots, f_s \rangle$ (see Section 21.7). Secondly, *Hilbert's irreducibility theorem* (1892), stating that for an irreducible polynomial $f \in \mathbb{Q}[x, y]$, the univariate polynomial $f(x, a) \in \mathbb{Q}[x]$ is irreducible for "most" $a \in \mathbb{Z}$. This sounds useful for reducing bivariate to univariate factorization. Unfortunately, no efficient versions of "most" are known, but, fortunately, such versions are known for reducing from many to two variables (Section 16.6).

Hilbert became a professor at the university of Göttingen in 1895. Under his leadership and that of Felix Klein, its fame, established by Gauß, as a center for mathematics kept growing. Among their famous colleagues were Hermann Minkowski, Ernst Zermelo, Constantin Carathéodory, Emmy Noether, Hermann Weyl, Carl Runge, Richard Courant, Edmund Landau, Alexander Ostrowski, Carl Ludwig Siegel, and Bartel van der Waerden, who based his *Modern Algebra* (1930b, 1931) on Emmy Noether's Göttingen lectures.

Hilbert's *Zahlbericht*[2], commissioned by the *Deutsche Mathematiker-Vereinigung*[3], gave a rich overview of the state of algebraic number theory and led him to a vast and elegant generalization of Gauß' quadratic reciprocity law and to the *Hilbert class field theory*.

His next area of work culminated in the booklet *Grundlagen der Geometrie*[4], where he laid down the basic properties that a "nice" system of axioms should have: soundness, completeness, and independence.

[1] Nullstelle = root
[2] Report on [the theory of] numbers
[3] German Mathematical Society
[4] Foundations of Geometry

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Then came what turned out to be his most influential "work": his talk on August 8, 1900, at the International Congress of Mathematicians in Paris (Hilbert 1900). He began with: *Wer von uns würde nicht gern den Schleier lüften, unter dem die Zukunft verborgen liegt, um einen Blick zu werfen auf die bevorstehenden Fortschritte unserer Wissenschaft und in die Geheimnisse ihrer Entwicklung während der künftigen Jahrhunderte!*[5], and ended with the list of the 23 *Hilbert problems*. As intended, this set of problems shaped the mathematics of the next century, and those who contributed to a solution would be said to belong to the "honors class" of mathematicians.



Hilbert liked lecturing, and excelled at it. He usually prepared only an outline of his lecture and filled in the details in front of the students—so he got stuck and confused everybody at times, but "a third of his lectures were superb". He was *Doktorvater* to the impressive number of 69 doctoral students, and this *Hilbert school* spread his approach to mathematics around the world.

Hilbert could be funny and entertaining at social events, loved dancing, and was a successful charmer of the ladies. His unprejudiced and liberal thinking led to a clash with the German authorities when he refused to sign, at the beginning of World War I, a declaration supporting the Kaiser and his government. At the beginning of the next German catastrophe, the Nazis forced in 1933 almost all Jewish professors out of their positions (and brutally worse was to come). Constance Reid (1970) relates in her wonderful biography how the Nazi minister of education said to Hilbert at a banquet in 1933 that mathematical life in

[5] Who of us would not be glad to lift the veil behind which the future lies hidden, to cast a glance at the next advances of our science and at the secrets of its development during future centuries!

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Göttingen probably had not suffered from being freed of Jewish influence. Hilbert's reply: *Jelitten? Das hat nicht jelitten, das jibt es nicht mehr.*[6]

After work on the Dirichlet Principle, Waring's Problem, the transcendence of e and π, integral equations, Hilbert spaces (*spectral theory*), calculus of variations, and a less successful attempt at laying the foundations of modern physics, Hilbert returned to logic and the foundations of mathematics in the 1920s. The 19th century philosopher Emil du Bois-Reymond had pointed to the limits of our understanding of nature: *ignoramus et ignorabimus*[7]. Hilbert was strongly opposed to this scepticism (*in der Mathematik gibt es kein ignorabimus*[8]) and set himself the goal of formalizing mathematics in a symbolic way, as pioneered by Gottlob Frege, and Bertrand Russell and Alfred North Whitehead. Alas, Hilbert's program was proved to be infeasible on this point by Kurt Gödel and Alan Turing; see Section 14.6 for the interesting juxtaposition of Hilbert's belief and a precocious undecidability result in polynomial factorization by van der Waerden. Although that particular goal of Hilbert's turned out to be unattainable, the ideas he introduced into proof theory and symbolic logic are alive and well today; see Section 24.1 for a small example. In fact, modern programming languages realize, in some sense, Hilbert's program of formalizing mathematics and science.

In the last decade of Hilbert's life, his health—including his mental facilities—deteriorated, and he led a secluded life. He died in February 1943, of the long-term effects of a fall. By then, the war had his country in its grip, and only a miserable procession of a dozen people accompanied the great mathematician on his last trip.

[6] Suffered? It did not suffer, it does not exist any more.
[7] we do not know and we shall not know
[8] in mathematics there is no ignorabimus [quoted from his 1900 lecture.]

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Tant que l'Algèbre et la Géométrie ont été séparées, leurs progrès
ont été lents et leurs usages bornés; mais lorsque ces deux sciences
se sont réunies, elles se sont prêté des forces mutuelles
et ont marché ensemble d'un pas rapide vers la perfection.[1]

Joseph Louis Lagrange (1795)

À la vérité le travail qu'il faudra faire pour trouver
ce diviseur, sera de nature, dans plusieurs cas,
à rebuter le Calculateur le plus intrépide. [. . .] Dans un travail
aussi long que l'est souvent celui de l'élimination,
il n'est pas inutile de multiplier les méthodes
sur lesquelles les Calculateurs peuvent porter leur choix.[2]

Étienne Bézout (1764)

The theory of modular systems is very incomplete and offers
a wide field for research. The object of the algebraic theory is
to discover those general properties of a module [polynomial ideal]
which will afford a means of answering the question
whether a given polynomial is a member of a given module or not.
Such a question in its simpler aspect is of importance in Geometry
and in its general aspect is of importance in Algebra.

Francis Sowerby Macaulay (1916)

[1] As long as algebra and geometry proceeded separately, their progress was slow and their application limited; but when these two sciences joined forces, they mutually strengthened each other, and marched together at a rapid pace toward perfection.

[2] Actually the effort required to find this divisor will, in several cases, be so large as to discourage the most intrepid Computer. [. . .] In an undertaking that is as hard as elimination often is, it is not useless to multiply the methods between which Computers can make their choice.

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